

Albert Street Cycling Counter

Data Quality Assessment

Counter 9077 | Team A cycling-counter workflow

1. Purpose

This report records the data-quality checks completed for the Albert Street bicycle counter (SITE_ID 9077) before the before-and-after analysis. The assessment focuses on data availability, missing periods, duplicate records, their likely source pattern, and whether exact duplicate removal materially changes the analytical result.

2. Data availability and missing periods

The availability checks identified two missing periods for Albert Street. The main gap covers 5 August 2019 to 19 January 2020 (168 days) and occurs in the before-intervention period. A second gap covers 26 December 2022 to 31 December 2022 (6 days) at the end of the expected study period. Because the 168-day gap cannot be reliably reconstructed from neighbouring observations, the analysis uses observed days only and reports the missing period as an important limitation. Missing days are not treated as zero and are not linearly interpolated.

3. Duplicate-record investigation

The extraction-stage duplicate report contained 1,996 Albert Street rows requiring investigation. Tracing repeated observations back to their source ZIP and CSV files showed that the duplication does not arise from a single pattern.

Source pattern	Duplicate groups	Rows in duplicate report
Across different ZIP files	572	1,144
Within the same CSV file	243	606
Across different CSV files in one ZIP	123	246

The largest category was observations appearing across different ZIP archives. This is consistent with overlapping source archives or repeated exports in the supplied data. Other repeated observations occurred within a single CSV or across different CSVs packaged in the same ZIP. The evidence therefore suggests that the duplicate issue is primarily associated with the supplied source files and their overlap, rather than one simple extraction-pipeline error.

4. Duplicate sensitivity analysis

A separate sensitivity test compared the original Albert Street raw records with a copy in which exact duplicate rows were removed. This cleaning step affected 202 observed days and removed 363 exact duplicate detections. The maximum reduction on a single day was 13 detections.

Method	Before average	After average	Percentage change
Original raw data	939.88	611.14	-34.98%

Exact duplicates removed	939.33	610.96	-34.96%
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Removing exact duplicates changed the estimated before-and-after percentage change by only 0.02 percentage points. The before-period average changed by approximately 0.56 detections per observed day and the after-period average by approximately 0.18. The duplicate-removal decision therefore has a negligible effect on the overall before-and-after result.

5. Data-quality decisions

- Retain the original source data unchanged.
- Record both missing periods explicitly: 5 August 2019 to 19 January 2020 (168 days) and 26 December 2022 to 31 December 2022 (6 days). Analyse observed days only; do not interpolate the 168-day gap or treat missing dates as zero.
- Treat exact duplicate records as a documented data-quality issue.
- For a cleaned analytical dataset, exact duplicates may be removed because they are repeated identical records; preserve the unmodified raw data separately.
- Keep the duplicate-source investigation and sensitivity analysis as evidence for the cleaning decision.
- Report that exact duplicate removal does not materially alter the Albert Street before-and-after result.

6. Limitations and interpretation

The duplicate sensitivity result does not remove the broader limitations of the before-and-after analysis. The 168-day gap from 5 August 2019 to 19 January 2020 is particularly important because it occurs in the before-intervention period. The 6-day gap from 26 to 31 December 2022 should also be documented, although it occurs at the end of the after-intervention period and is less substantial. The comparison may also be influenced by seasonal effects, COVID-19 disruptions, and other external changes. The observed reduction in bicycle detections should therefore be reported as an association between the available before and after observations, not as proof that the intervention alone caused the change.

7. Recommended project record

Keep this assessment with the cycling-counter workflow and retain the generated quality tables, duplicate-source outputs, and sensitivity-analysis outputs. This provides a clear record of what was checked, what cleaning decision was made, and why the final analysis can be reproduced by later project teams.

Status: Albert Street data-quality assessment completed for the checks described above.